*Bioinformatics Advances*, 2026, pp. 1–9

**doi: DOI HERE**
Advance Access Publication Date: Day Month Year
Original Paper

ORIGINAL PAPER

# Benchmarking the accuracy, robustness and reproducibility of spatial domain identification methods

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FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

## Abstract

**Motivation:** New methods for spatial domain identification are usually compared by small differences in adjusted Rand index (ARI), mostly on the twelve human DLPFC Visium sections. Most of them obtain domains by clustering a learned embedding, typically with a single run of a Gaussian mixture model (mclust, EEE). How much of the reported differences is due to the encoder and how much to this final clustering step is rarely measured.
**Results:** We evaluated 13 published methods (HiSTaR, GraphST, BANKSY, SpaGCN, STAGATE, SEDR, SpaceFlow, NichePCA, CCST, Spatial-MGCN, DeepST, BASS and BayesSpace) and three training-free graph-diffusion baselines on 20 annotated sections from three technologies (10x Visium, STARmap, MERFISH). With its official code, HiSTaR reached a median ARI of 0.51 on DLPFC instead of the reported 0.65. On 200 fixed embeddings from 10 methods, refitting the same mixture model from different random starts changed ARI by 0.17 on average (0.09–0.23 depending on the method) and by up to 0.60; selecting the best of 20 fits by ARI would inflate results by 0.07. R mclust is deterministic for a given input, but merely permuting the order of the spots changed its ARI by up to 0.37, and adding noise at 1% of the feature scale by up to 0.56. Selecting fits by likelihood did not resolve this: matched 20-start clustering helped 6 of 12 methods and hurt 5. The highest mean ARI was reached by linear diffusion on DLPFC, BANKSY $\lambda = 0.8$ on STARmap and BASS on MERFISH; the training-free linear diffusion baseline was not significantly different from the best methods and significantly better than 17 of 24 comparisons. In addition, the released code of one method selects its reported epoch by agreement with the labels, which raised its mean ARI by 0.147. We give reporting recommendations for benchmarks.
**Availability and implementation:** Code, per-run results and figures: [TODO: GitHub URL and Zenodo DOI]. All public data sources are listed in the Data availability section.

**Key words:** spatial transcriptomics, spatial domains, clustering, reproducibility, benchmarking

## Introduction

Spatial transcriptomics measures gene expression at thousands of locations in a tissue section while retaining their coordinates. Identifying *spatial domains*, regions with coherent expression such as cortical layers, is one of the first analyses of such data, and many methods have been proposed for it, from Bayesian models [Zhao et al., 2021] and neighbourhood-augmented features [Singhal et al., 2024] to graph neural networks [Hu et al., 2021, Dong and Zhang, 2022, Long et al., 2023] and hierarchical graph variational autoencoders such as HiSTaR [Yu et al., 2025].

Most of these methods are evaluated on the twelve manually annotated human dorsolateral prefrontal cortex (DLPFC) sections [Maynard et al., 2021] and report median ARI improvements of a few hundredths over previous methods. Many share the same last step: the learned embedding is clustered once with mclust's EEE Gaussian mixture [Scrucca et al., 2016]. Benchmarks have found that rankings depend strongly on the dataset [Yuan et al., 2024, VERIFY, 2025a], that deep methods vary with the random seed [VERIFY, 2025a], that replacing each method's clustering algorithm changes results [Hu et al., 2024], and that preprocessing and clustering choices often matter more than architectural novelty [VERIFY, 2026]. Simple training-free pipelines are also competitive [VERIFY, 2025b, Do et al., 2026].

We complement these studies with a direct measurement of the variance contributed by the clustering step itself, on a larger set of methods. We ask: (1) do HiSTaR's published DLPFC results reproduce with the official code; (2) on a *fixed* embedding, how much does ARI change with the initialisation of the mixture model, and with perturbations that leave the data essentially unchanged (spot order, negligible noise) under R mclust; (3) does selecting fits by likelihood help; and (4) how do 13 published methods and training-free baselines compare

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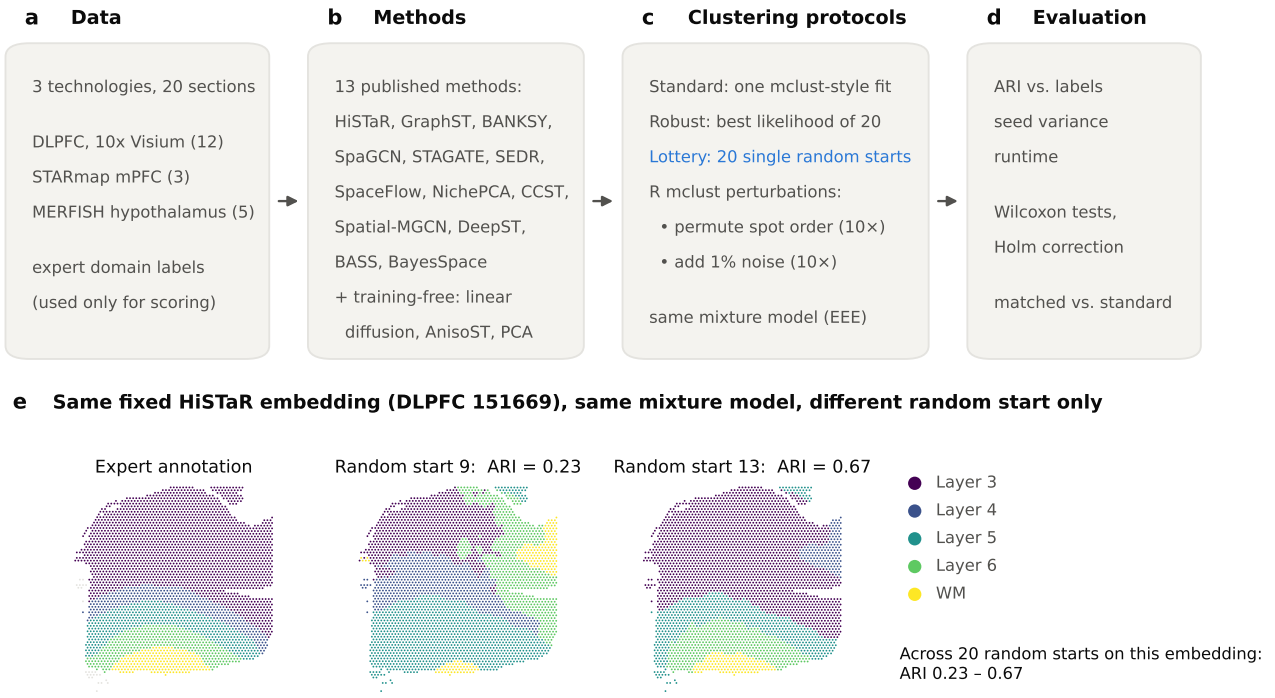



**Fig. 1.** Study design. **a**–**d**: data, methods, clustering protocols and evaluation. **e**: a real example of the clustering lottery. One fixed HiSTaR embedding of DLPFC section 151669 was clustered with the same tied-covariance Gaussian mixture from two different random starts; only the initialisation differs, yet ARI is 0.23 in one run and 0.67 in the other.

across three technologies, and does the ranking depend on how the embeddings are clustered? The study design is summarised in Fig. 1.

## Materials and methods

### Data

**DLPFC (10x Visium).** Twelve sections from three donors [Maynard et al., 2021]; seven annotated layers (five for donor 2), `layer_guess_reordered`. **STARmap mouse medial prefrontal cortex.** Three sections (about 1050 cells, 166 genes) with four annotated layers (L1, L2/3, L5, L6) [Wang et al., 2018]. **MERFISH mouse hypothalamus.** Animal 1, five sections from Bregma −0.04 to −0.24 (about 5500 cells, 155 genes) with eight annotated regions [Moffitt et al., 2018]. The STARmap and MERFISH annotations are those distributed with the BASS analyses [Li and Zhou, 2022]. In total, 20 sections from three technologies.

### Preprocessing

DLPFC: genes in fewer than 50 spots or with fewer than 10 counts were removed, 2000 highly variable genes were selected (Seurat v3 flavour), counts were library-size normalised, log-transformed and scaled, and reduced to 200 principal components (PCs). STARmap and MERFISH (targeted panels): all genes, same normalisation, 50 PCs. All methods received the same input where they accept one; BANKSY used the log-normalised genes, and SpaGCN, GraphST and the methods added later (below) their own documented preprocessing of the same raw counts. The true number of domains was given to every method, and unannotated spots were excluded from evaluation.

### Methods

- **HiSTaR** [Yu et al., 2025]: PyTorch port of the official code with the published defaults (cross-level weight 0.3, 200+200 epochs).
- **GraphST** [Long et al., 2023]: official package and tutorial pipeline (600 epochs, 20-PC embedding, R mclust EEE; with and without its label refinement). Because of its CPU cost, GraphST was run with one seed on DLPFC and three seeds on STARmap and MERFISH.
- **BANKSY** [Singhal et al., 2024]: official `pybanksy`, 18 neighbours, $m_{\max} = 1$, $\lambda \in \{0.2, 0.8\}$, 20 PCs.
- **SpaGCN** [Hu et al., 2021]: official tutorial pipeline without histology ($p = 0.5$, Louvain resolution search to the true number of domains, 200 epochs).
- **Methods added in the extended study**, each run with its official code and tutorial settings on one CPU thread: STAGATE [Dong and Zhang, 2022] (R mclust), SEDR [VERIFY, 2024] (R mclust), SpaceFlow [VERIFY, 2022b] (Leiden with a search for the true number of domains), NichePCA [VERIFY, 2025b] (Leiden, same search), CCST [VERIFY, 2022a] and Spatial-MGCN [VERIFY, 2023] (both $k$-means), DeepST [VERIFY, 2022c] (Leiden with refinement), BASS [Li and Zhou, 2022] and BayesSpace [Zhao et al., 2021] (Bayesian models that cluster internally; BayesSpace requires a spot lattice and was run on the Visium sections only). NichePCA, SEDR, SpaceFlow and DeepST were run with three seeds per section; STAGATE, CCST, Spatial-MGCN and BASS with one seed (seed 0)

and BayesSpace once per section. Deviations from the tutorials (for example the neighbourhood radius, expressed in spot spacings so that it transfers across technologies) are documented in the code repository. Spatial-MGCN's released test script reports the epoch with the highest ARI against the annotation; in the benchmark we report its final epoch (label-free) and give the label-selected value separately.

- **Training-free baselines.** *Linear diffusion*: the top 30 PCs are smoothed on the spatial 6-nearest-neighbour graph, $H^{(t+1)} = (1-\alpha)H^{(t)} + \alpha D^{-1}AH^{(t)}$, $\alpha = 0.5$, $T = 10$. *AnisoST*: the same with edge weights recomputed at every step, $w_{ij}^{(t)} = \exp(-\|h_i^{(t)} - h_j^{(t)}\|^2/\kappa^{(t)})$ (Perona–Malik-type edge-preserving diffusion [Perona and Malik, 1990]; $\kappa^{(t)}$ the median squared edge difference); a similar idea has been used inside a deep model [Wang et al., 2024]. *PCA* without smoothing is the non-spatial reference. All baseline settings were fixed on DLPFC donor 1 before any other data were analysed, and none were tuned on STARmap or MERFISH. All methods were run with default or published settings; none were tuned per dataset (equal tuning budgets are a topic for future work).

### Clustering protocols

*Standard*: each method's usual clustering. This is R mclust for GraphST, STAGATE and SEDR, a 3-start scikit-learn GMM [Pedregosa et al., 2011] with a fixed seed for HiSTaR, BANKSY and the baselines, Leiden or $k$-means for the other methods, and the internal model for SpaGCN, BASS and BayesSpace. *Robust*: the method's embedding clustered with a tied-covariance Gaussian mixture (the mclust EEE model) with 20 random starts, keeping the fit with the highest log-likelihood (label free); available for every method that outputs an embedding. *Lottery*: one seed-0 embedding per section and method, clustered 20 times with a single random start each (10 methods with embeddings). *mclust perturbations*: R mclust (version 6.0.1, default hierarchical initialisation) on the same embedding (i) unchanged, (ii) after random permutation of the rows, 10 times, and (iii) after adding Gaussian noise with 1% of each feature's standard deviation, 10 times (10 methods in the extended set, plus HiSTaR, GraphST and AnisoST).

### Evaluation and statistics

ARI [Hubert and Arabie, 1985] against the manual annotation; NMI gave the same conclusions (code repository). For every (method, section) we average over seeds. Methods were compared with two-sided Wilcoxon signed-rank tests over sections, with Holm correction [Holm, 1979] over the family of 24 comparisons of linear diffusion with each method and clustering protocol. Run time and peak memory were measured for a subset of methods in a standardised runner (process-tree sampling, one CPU thread per job).

## Results

### HiSTaR's published DLPFC results did not reproduce

With the official code and defaults, HiSTaR reached a median ARI over the twelve DLPFC sections of 0.509 (mean 0.491, 95% bootstrap CI 0.454–0.525) with the 3-start GMM, and 0.449 (mean, seed 0) when its embedding was clustered with R mclust as in the official pipeline, compared with a reported median of 0.65 [TODO: verify the reported values (0.65 median; 0.718 on 151673) against the HiSTaR article]. Fixing a bug that applies random masking at inference did not change accuracy (0.487). Possible causes include per-section tuning of the loss weights (described in the HiSTaR supplement), unpublished preprocessing and the choice of seed; the following sections show that the clustering step alone produces differences of this size. [TODO: Report the outcome of contacting the authors.]

### On a fixed embedding, the clustering step alone moves ARI by up to 0.60

Figure 2 shows the ARI of 20 single-start fits of the same mixture model on each fixed embedding. Across the 200 embeddings of 10 methods, the ARI range within an embedding was 0.17 on average and reached 0.60 (SEDR, dlpfc 151672). The effect depended on the method: the mean range was 0.09 for SpaceFlow and 0.23 for HiSTaR (Fig. 3); for HiSTaR and AnisoST it was 0.23 and 0.20, whereas GraphST's 20-dimensional PCA embeddings were more stable (0.09). The best of the 20 fits by ARI exceeded the average single fit by 0.07: choosing a clustering run by its agreement with the labels, explicitly or by trying a few seeds, would inflate a reported result by about as much as the gap between most published methods.

### R mclust is deterministic but not stable

mclust initialises EM by model-based hierarchical agglomeration, so one call on a fixed embedding always returns the same partition. That partition, however, depended on details that carry no information (Fig. 4). Over 200 embeddings, randomly permuting the order of the spots changed ARI by 0.09 on average within an embedding (up to 0.37); on DLPFC the mean was 0.11 and on MERFISH 0.10. In 21% of the permutations ARI moved by more than 0.05, and for 40% of the embeddings the range exceeded 0.1. Adding noise at 1% of each feature's standard deviation had similar effects (mean range 0.12, up to 0.56; 0.21 on STARmap). Over these embeddings, mclust's default fit had a mean ARI of 0.40 against 0.42 for the best-likelihood fit of 20 random starts, and its log-likelihood was at least as high in only 40% of cases.

### Choosing fits by likelihood is not a remedy

A natural fix is to run many starts and keep the most likely fit. It helped some methods and hurt others. Replacing each method's standard clustering by this matched protocol raised mean ARI for 6 of 12 methods and lowered it for 5. For example, it raised HiSTaR's mean ARI by $+0.053$ (14/20 sections, $p = 0.004$) and GraphST's by $+0.025$ (13/20, $p = 0.114$), but changed SEDR's by $-0.092$ and Spatial-MGCN's by $-0.037$. The reason is visible in Fig. 2: likelihood and ARI were positively related on DLPFC (median Spearman $\rho$ 0.40) but not on MERFISH (median $\rho$ $-0.16$; negative for 7 of 10 methods), where the best-fitting mixture can separate cell types rather than anatomical regions. Better optimisation of the mixture model is therefore not the same as better domains.

### Under matched clustering, no deep encoder outperformed a training-free baseline

Table 1 and Fig. 5 compare all methods under both protocols. Across all methods and both protocols, the highest mean ARI was reached by linear diffusion on DLPFC (0.558), BANKSY $\lambda = 0.8$ on STARmap (0.740) and BASS on MERFISH (0.582).

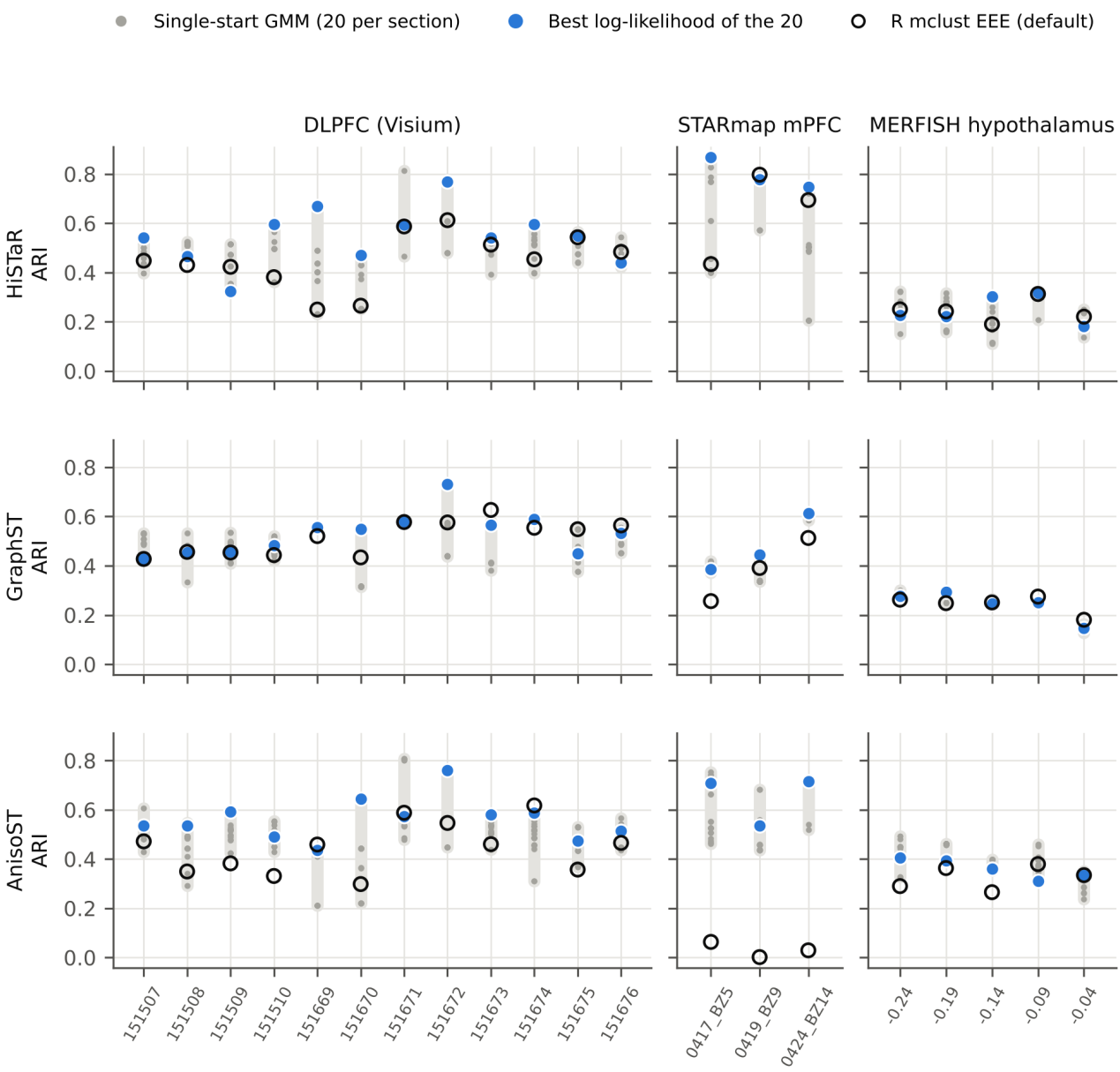



**Fig. 2.** The clustering lottery (HiSTaR, GraphST and AnisoST; the other methods are in Fig. 3 and the repository). For each section, one fixed embedding was clustered 20 times with a single-start tied-covariance GMM (grey dots; bar = range). Blue: the fit with the highest log-likelihood (label-free). Open circle: R mclust with default settings on the same embedding.

Linear graph diffusion had a mean ARI of 0.558, 0.705 and 0.497 on the three datasets (average over datasets 0.587, against 0.594 for BASS and 0.554 for BANKSY $\lambda = 0.8$). Over the 20 sections it had a higher mean ARI than most competitors: the difference was significant after Holm correction in 17 of the 24 comparisons, including HiSTaR with standard clustering (+0.126, 17/20 sections, Holm-adjusted $p = 0.009$), GraphST with its own mclust clustering (+0.135, 14/20, adjusted $p = 0.030$), SpaGCN (+0.234, 20/20, $p < 0.001$), NichePCA (+0.150, adjusted $p = 0.004$), SpaceFlow (+0.186, $p < 0.001$), CCST (+0.174, $p < 0.001$), DeepST (+0.177, $p < 0.001$) and Spatial-MGCN (+0.223, $p < 0.001$). The differences to BANKSY $\lambda = 0.8$ (+0.051, adjusted $p = 0.120$), AnisoST (+0.050, $p = 0.120$), SEDR (+0.065, $p = 0.120$), BASS (+0.021, $p = 0.648$), STAGATE and HiSTaR with robust clustering (+0.065 and +0.074, adjusted $p = 0.120$ and $p = 0.120$) were not significant: these methods are statistically indistinguishable from the baseline with 20 sections, which is not evidence that they are equivalent. The ranking of the deep methods relative to the baselines changed with the clustering protocol (Fig. 5). The deep methods were also the least stable: HiSTaR's seed-to-seed standard deviation of ARI was 0.047 on DLPFC and 0.18 on STARmap, against 0.003–0.04 for the training-free baselines. Run time and memory differed by orders of magnitude between methods (Table 2); in a separate feasibility test, one DLPFC section took tens of minutes on

**Table 1.** Mean ARI over sections. Standard: each method's usual clustering (R mclust for GraphST, STAGATE, SEDR; 3-start GMM for HiSTaR, BANKSY and the baselines; Leiden or $k$-means for the other methods; SpaGCN, BASS and BayesSpace cluster internally). Robust: 20-start GMM, best likelihood, on the same embeddings. Seeds per section: 3 for HiSTaR, BANKSY, the baselines, NichePCA, SEDR, SpaceFlow, DeepST and GraphST on STARmap and MERFISH; 1 for GraphST on DLPFC, STAGATE, CCST, Spatial-MGCN, BASS and BayesSpace. †Training-free; all baseline settings fixed on DLPFC donor 1.

| | DLPFC (Visium) | | STARmap mPFC | | MERFISH hypothalamus | |
|---|---|---|---|---|---|---|
| | Standard | Robust | Standard | Robust | Standard | Robust |
| Linear diffusion† | – | 0.558 | – | 0.705 | – | 0.497 |
| AnisoST† | 0.526 | 0.551 | 0.564 | 0.622 | 0.377 | 0.363 |
| PCA, non-spatial† | – | 0.409 | – | 0.294 | – | 0.130 |
| BANKSY $\lambda = 0.8$ | 0.494 | 0.518 | 0.740 | 0.694 | 0.428 | 0.410 |
| BANKSY $\lambda = 0.2$ | 0.514 | 0.518 | 0.502 | 0.529 | 0.211 | 0.222 |
| HiSTaR | 0.491 | 0.544 | 0.593 | 0.658 | 0.221 | 0.264 |
| GraphST | 0.514 | 0.531 | 0.388 | 0.484 | 0.251 | 0.254 |
| SpaGCN | 0.422 | – | 0.267 | – | 0.151 | – |
| STAGATE | 0.497 | 0.548 | 0.479 | 0.572 | 0.368 | 0.342 |
| SEDR | 0.515 | 0.551 | 0.593 | 0.236 | 0.406 | 0.167 |
| SpaceFlow | 0.312 | 0.326 | 0.628 | 0.668 | 0.389 | 0.260 |
| NichePCA | 0.321 | 0.506 | 0.692 | 0.604 | 0.475 | 0.403 |
| CCST | 0.437 | 0.359 | 0.505 | 0.435 | 0.212 | 0.157 |
| Spatial-MGCN | 0.418 | 0.391 | 0.303 | 0.245 | 0.182 | 0.133 |
| DeepST | 0.459 | 0.435 | 0.368 | 0.409 | 0.231 | 0.163 |
| BASS | 0.484 | – | 0.716 | – | 0.582 | – |
| BayesSpace | 0.437 | – | – | – | – | – |

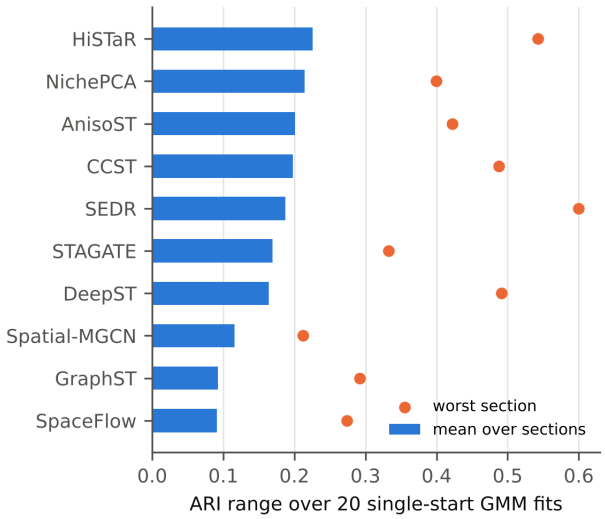



**Fig. 3.** ARI range over 20 single-start GMM fits of one fixed embedding, by method: mean over the sections (bars) and the worst section (dots).

one CPU thread for most of the neural methods and about 50 minutes for BayesSpace.

## One released script selects its reported epoch by agreement with the labels

[TODO: Contact the Spatial-MGCN authors and report their response before including this subsection.] Spatial-MGCN's released test scripts (`DLPFC_test.py`, `HBC_test.py`) compute ARI against the annotation at every training epoch and keep the best epoch for reporting. Applying the same selection rule in our runs (seed 0, 20 sections), this label-selected epoch had a mean ARI of 0.551 on DLPFC, 0.687 on STARmap and 0.221 on MERFISH, against 0.418, 0.303 and 0.182 for the final epoch used in our benchmark, i.e. an inflation of 0.147 on average (0.38 on STARmap). Selecting a run, epoch or seed by agreement with the labels is a form of label leakage that makes a method look competitive with the training-free baselines (0.551 against 0.558 on DLPFC) when its label-free result is not.

**Table 2.** Time and memory of a whole job (data loading, embedding and clustering) in a standardised runner, one CPU thread, for the methods run in it; the other methods were not measured there.

| Method | Median time per section (s) | Peak memory (GB) |
|---|---|---|
| Linear diffusion | 24 | 1.4 |
| AnisoST | 23 | 1.4 |
| PCA | 25 | 1.4 |
| BANKSY | 22 | 1.9 |
| HiSTaR | 119 | 1.6 |
| SEDR | 104 | 2.4 |
| NichePCA | 56 | 7.2 |

## Edge preservation helps at boundaries on Visium but not on single-cell data

On DLPFC, AnisoST matched linear diffusion in ARI (0.551 vs 0.558) and improved accuracy on spots at layer boundaries by 0.025 (9/12 sections, $p = 0.034$, uncorrected) without changing interior accuracy (Supplementary Fig. S2). On MERFISH it was clearly worse (0.363 vs 0.497): at single-cell resolution neighbouring cells often differ by cell type, so an edge-preserving rule blocks the smoothing that is needed to reveal regions. We report this as a negative result for the edge-preserving variant; the simpler linear diffusion was the more robust baseline.

# Discussion

The final clustering step of spatial domain methods is a substantial and under-reported source of variance. On fixed

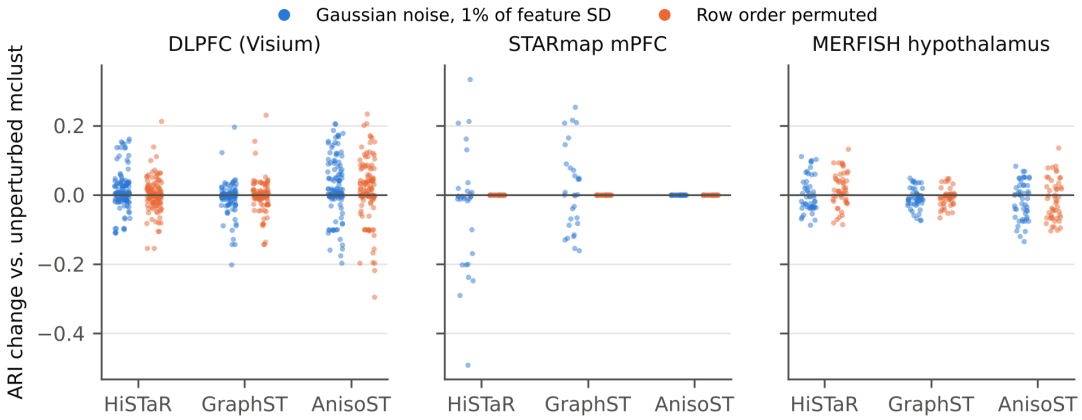



**Fig. 4.** Change in ARI of R mclust (EEE, default initialisation) when the same embedding is perturbed by permuting the spot order (orange) or adding noise at 1% of each feature's standard deviation (blue); 10 perturbations per section and embedding.

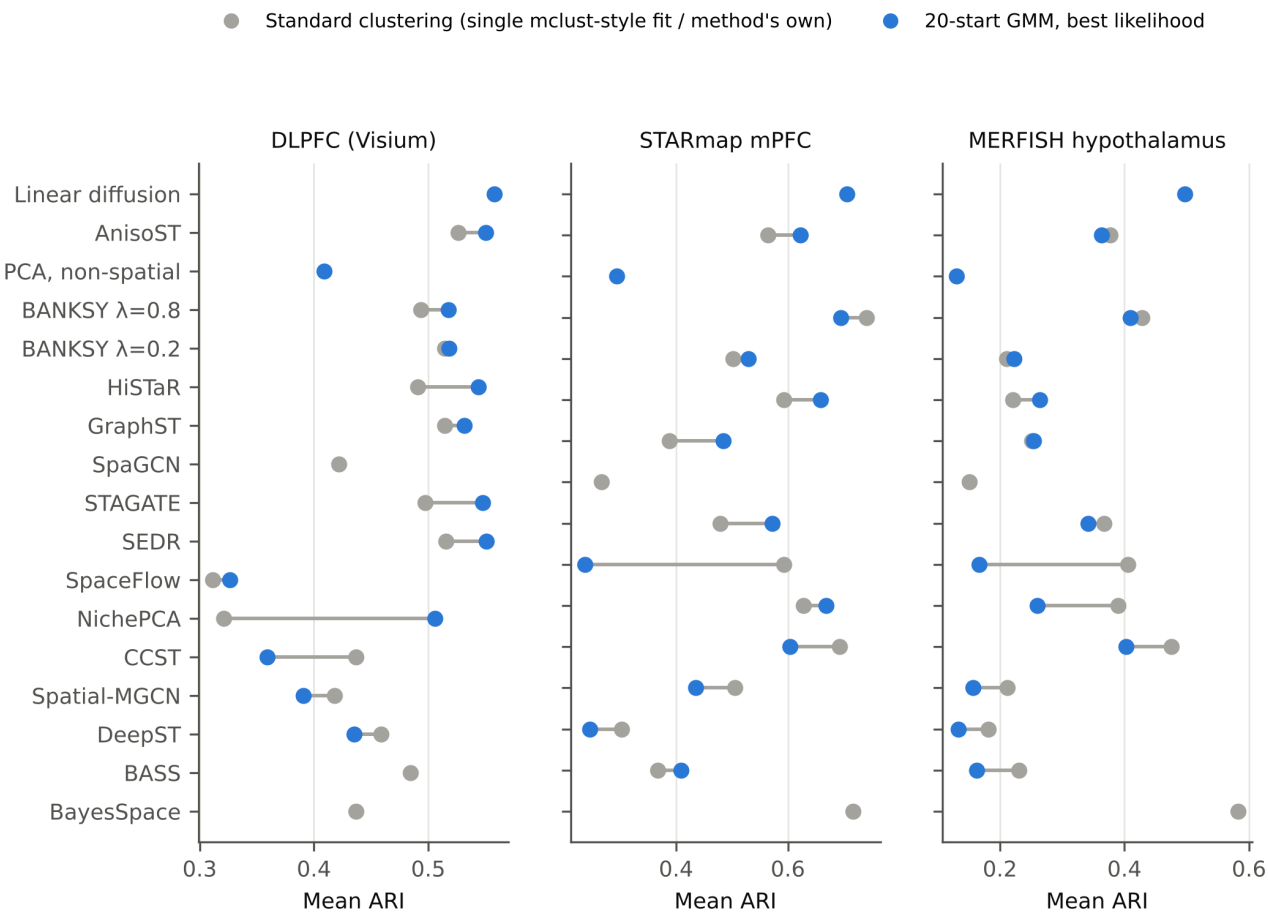



**Fig. 5.** Mean ARI per dataset with standard clustering (grey) and the 20-start, best-likelihood GMM (blue) applied to the same embeddings.

embeddings from 10 methods and three technologies, random initialisation alone spanned 0.17 ARI on average, and the R implementation used by most published methods changed its answer when the spots were merely reordered. Better likelihood did not reliably mean better domains, and a training-free diffusion baseline matched or outperformed every deep encoder we tested. These results agree with recent large benchmarks [VERIFY, 2026, Hu et al., 2024, VERIFY, 2025a] and add a direct, quantitative measurement of clustering variance on fixed embeddings across a broader set of methods.

Recommendations.

(1) Report the clustering protocol: algorithm, implementation and version, initialisation, number of starts, seeds, and how they were chosen. (2) Never select seeds, epochs or runs by agreement with the labels. (3) Report distributions over training seeds *and* clustering runs (or input permutations), with per-section values. (4) Compare encoders under the same clustering, and include training-free baselines such as neighbourhood smoothing. (5) Report runtime, memory and seed variance alongside accuracy.

Limitations.

All methods were run with default or published settings (we did not tune any method and we could not reproduce HiSTaR's per-section tuning), so our results do not show what each method can reach with its authors' own tuning. Several methods were run with one seed, and BayesSpace on Visium only. The lottery and mclust experiments use one embedding per section and method. The STARmap and MERFISH annotations are coarse and the STARmap data contain only three small sections. Three technologies and 20 sections do not cover the breadth of spatial data (for example Slide-seq, Stereo-seq or Xenium). The number of domains was given to every method. Annotations may themselves have been derived with methods similar to the ones evaluated.

## Competing interests

No competing interest is declared.

## Author contributions statement

[TODO: CRediT statement.]

## Use of AI tools

[TODO: Describe per journal policy: code, experiments and a first draft of this manuscript were produced with the assistance of an AI coding assistant (Claude, Anthropic); all results were checked by the authors, who take full responsibility for the content.]

## Data availability

DLPFC: spatialLIBD (count matrices) and SEDR analyses repository (layer annotations). STARmap and MERFISH with domain annotations: BASS-Analysis repository [Li and Zhou, 2022], from the original studies [Wang et al., 2018, Moffitt et al., 2018]. Scripts that download and convert all data are provided with the code.

## Funding

[TODO: Funding.]

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## Supplementary figures

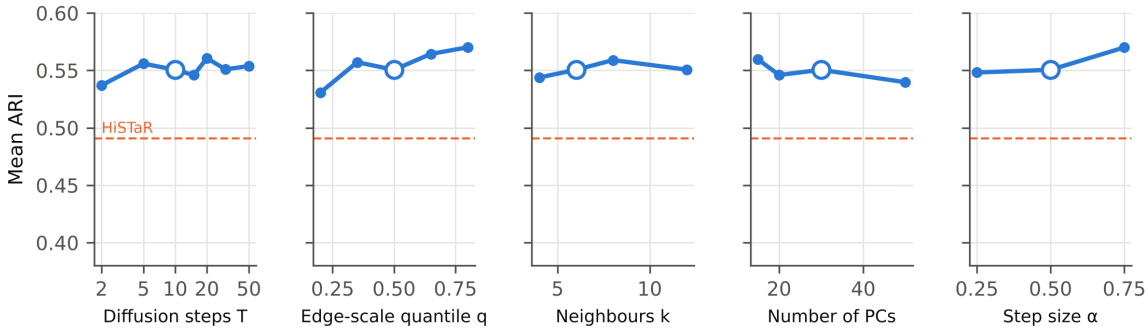



**Fig. S1.** DLPFC: mean ARI of AnisoST when one hyper-parameter is changed at a time; open circle = default; dashed = HiSTaR, standard clustering.

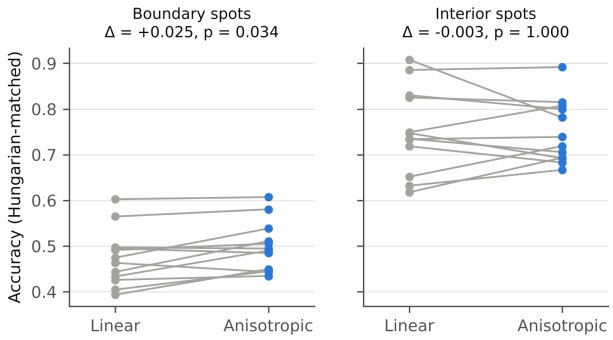



**Fig. S2.** DLPFC: accuracy on boundary and interior spots, linear vs. edge-preserving diffusion; one line per section.

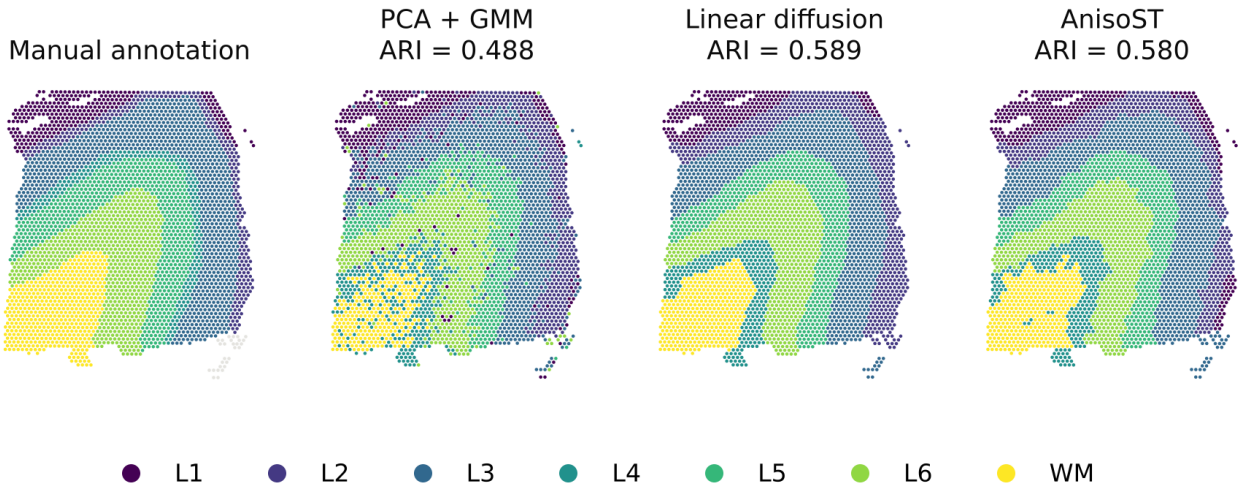



**Fig. S3.** DLPFC section 151673: manual annotation and domains from PCA + GMM, linear diffusion and AnisoST.